PHYSICS

Q1. (3) deviate towards the normal

Q2 (2) II

Q3. ~~(2) Atmospheric Refraction~~ (3) Scattering of light

Q4. (3) Ciliary muscle

Q5. (2)

Q6 the ratio of speed of light in vaccum to the speed of light in that medium is called refractive index of transparent medium

Given : $n = 1.5$

$v = 2 \times 10^8$ m/s

now, $n = c/v$

$c = \frac{n}{v}$

$c = \frac{1.5}{2 \times 10^8}$

$c = 3 \times 10^8$ m/s

$\therefore$ The speed of light in vacuum is 3×10^8 m/s

Q7. A fuse in a circuit prevents the damage to the
Ans appliances and the current due to overloading. The use of an electric fuse a possible damage by stopping the flow of unduly high electric current. The Joule heating that takes place in the fuse melts it to break the electric circuit. If a fuse with a defined rating replaced by one with larger rating, then the Joule heating fails to melt the fuse and hence, the circuit is not broken. This results in burning of appliances

Q8.
Ans. Given :-

normal eye near point u = -25cm

Defective nearer point v = -75cm

This problem is hypermetropia.

So, we use convex lens.

By using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-75} - \frac{1}{-25}$$

$$\frac{1}{f} = 37.5cm$$

The power is $P = \frac{1}{f}$

$$P = \frac{100}{37.5}$$

$$P = 2.66 D$$

∴ Hence the focal length is 37.5cm and the power is 2.66D

Ans 9 By $R = \frac{\rho L}{A}$

If $L' = 2L$

$A' = \frac{A}{2}$

Then $R' = \frac{\rho L'}{A'} = \frac{\rho(2L)}{(A/2)} = \frac{4\rho L}{A} = 4R$

So,

(i) The resistance increases by 4 times ~~of~~ if the length is doubled and the area is halved.

(ii) The resistivity remains unchanged. Resistivity is an ~~resistance will have~~ instrinstic property of a material and it is not scalable

~~Ans~~

C By Ohm's law $V = IR$

$\therefore R = \frac{V}{I}$ = slope of V-I graph

Therefore, the slope of the V-I graph represent the effective resistance

$R_{eq} = R_1 + R_2 + R_3 + R_4 + R_n$

∴ Therefore, the series combination will have the maximum resistance than the other resistors.

Ans: Given :- height of the object, $h_1 = 4cm$

Distance of the object from the mirror, $u = -25cm$

The focal length of the mirror, $f = -15cm$

By mirror formula

$$\cancel{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}} \qquad \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\cancel{\Rightarrow \frac{1}{-15} = \frac{1}{v} + \frac{1}{-25}} \qquad \frac{1}{v} + \frac{1}{-25} = \frac{1}{-15}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{-15} + \frac{1}{25} = \frac{(-25+15)}{375} = \frac{1}{-37.5cm}$$

$$\Rightarrow v = -37.5 cm$$

The image is formed object side 37.5 cm from the pole of the concave mirror.

Q. $\frac{h_2}{h_1} = \frac{-u}{v}$

where h_1 is the size of the image.

Now, $\frac{h_2}{4cm} = \frac{-(-37.5cm)}{(-25cm)}$

$$h_2 = -4 \times \frac{37.5}{25} = -6cm$$

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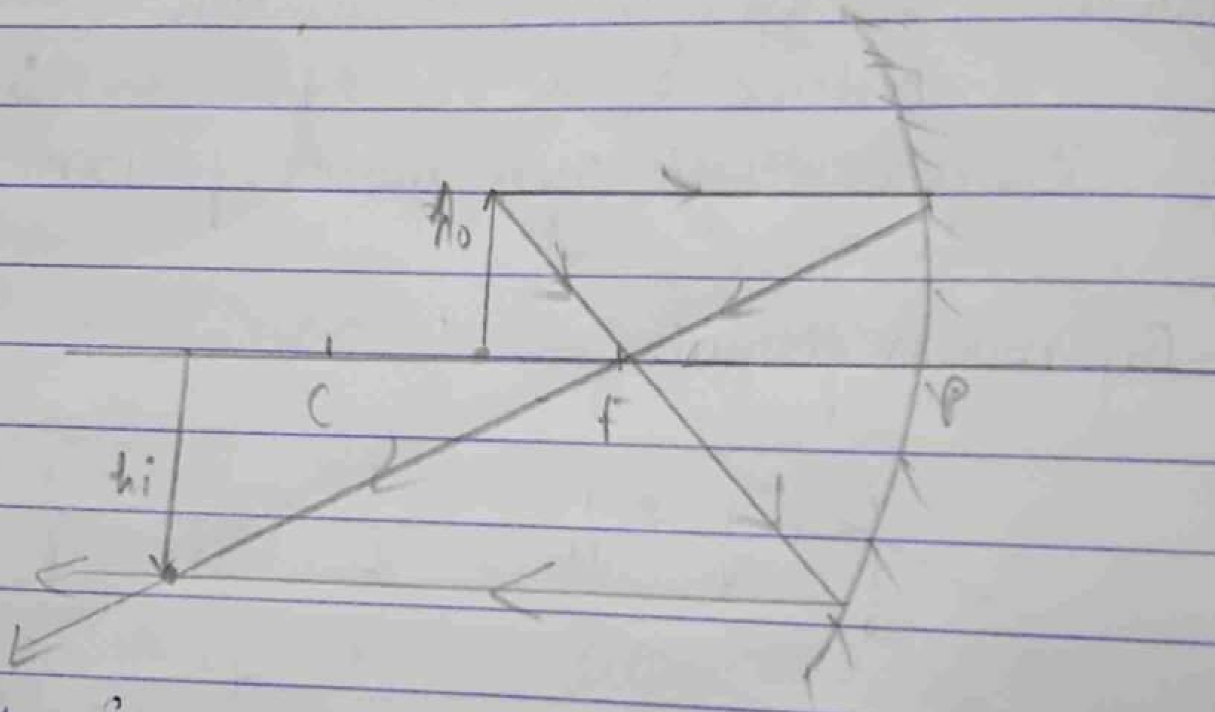
∴ Size of the image = 6 cm



So The image is enlarged and the nature is real and inverted